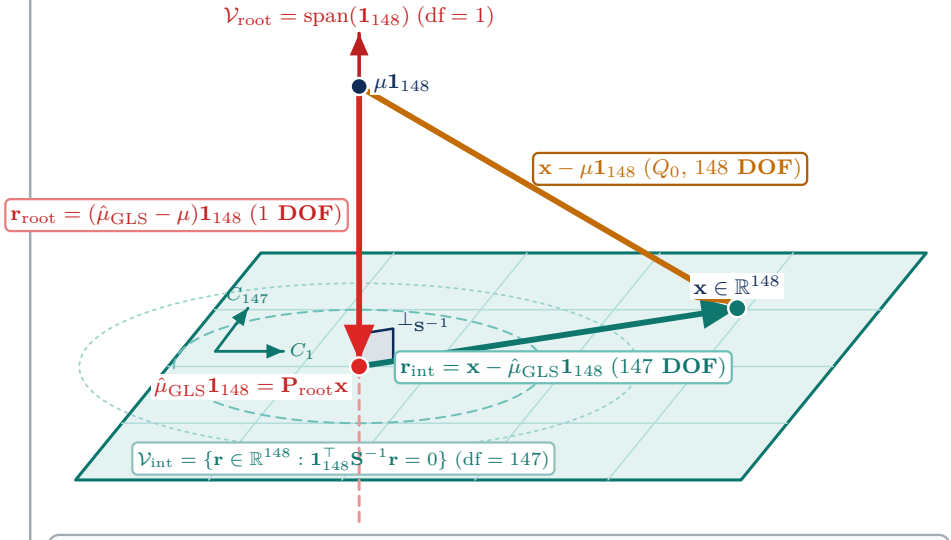
(A) $\mathcal{H} = \left(\mathbb{R}^{148}, \langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{S}^{-1}}\right), \quad \mathbb{R}^{148} = \mathcal{V}_{\text{int}}\ (147\ \textbf{DOF}) \oplus_{\mathbf{S}^{-1}} \mathcal{V}_{\text{root}}\ (1\ \textbf{DOF})$



$$Q_0(\mathbf{x}, \mu) = \|\mathbf{x} - \mu\mathbf{1}\|^2_{\mathbf{S}^{-1}} = \underbrace{(\mathbf{x} - \hat{\mu}_{\text{GLS}}\mathbf{1})^\top \mathbf{S}^{-1}(\mathbf{x} - \hat{\mu}_{\text{GLS}}\mathbf{1})}_{Q_{\text{internal}}(\mathbf{x})\ (\text{df}=147)} + \underbrace{\text{vec11} \cdot (\hat{\mu}_{\text{GLS}}(\mathbf{x}) - \mu)^2}_{Q_{\text{root}}(\mathbf{x},\mu)\ (\text{df}=1)}$$

(B) $\hat{\mu}_{\text{GLS}}(\mathbf{x}) \perp\!\!\!\perp Q_{\text{internal}}(\mathbf{x}) \implies (\hat{\mu}, \hat{\sigma}) \mid \mathbf{x}$

$\mathbf{x} \in \mathbb{R}^{148}$
(μ, σ) ẩn

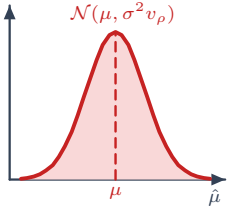
$\mathbf{P}_{\text{root}}$

$\mathcal{V}_{\text{root}}$ (df = 1 — **Location** $\hat{\mu} \perp \sigma$):

$\hat{\mu}_{\text{GLS}}(\mathbf{x}) = \mathbf{w}_{\text{GLS}}^\top \mathbf{x} = \frac{\mathbf{1}^\top \mathbf{S}^{-1}\mathbf{x}}{\text{vec11}}$

$\mathbf{x} + c\mathbf{1}_{148} \mapsto \hat{\mu}_{\text{GLS}} + c$

$\text{CRLB}(\hat{\mu}) = \sigma^2 v_\rho = 0.2022\,\sigma^2$



$\text{Cov}(\hat{\mu}_{\text{GLS}}\mathbf{1}_{148}, \mathbf{r}_{\text{int}}) = \mathbf{0} \iff \hat{\mu}_{\text{GLS}}(\mathbf{x}) \perp\!\!\!\perp Q_{\text{internal}}(\mathbf{x})$

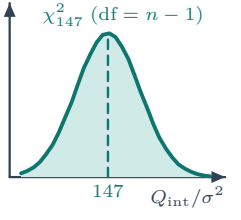
$\mathbf{I} - \mathbf{P}_{\text{root}}$

$\mathcal{V}_{\text{int}}$ (df = 147 — **Scale** $\hat{\sigma} \perp \mu$):

$Q_{\text{internal}}(\mathbf{x}) = \sum_{u=1}^{147} C_u^2, \quad C_u \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$

$\mathbf{x} + c\mathbf{1}_{148} \mapsto Q_{\text{internal}}(\mathbf{x})$

$\hat{\sigma}_{\text{REML}} = \sqrt{Q_{\text{internal}}(\mathbf{x})/147}$



phylolm & PhyloGNN_M5A: $\hat{\mu}(\mathbf{x}) = \mathbf{w}_\mu^\top \mathbf{H}_{\text{nodes}} \perp \sigma \oplus \hat{\sigma}(\mathbf{x}) = \gamma_{c_4} \sqrt{\mathbf{w}_\sigma^\top \mathbf{C}_{\text{int}}^{\odot 2}} \perp \mu$